

Corporate Finance (in lingua inglese) - A.A. 2024/2025 - Topic 5d and 8: NPV and other investment criteria

TEXT and SOLUTION (through the use of Excel)

Event Corporation is planning to set up a new gift shop. The firm will use a building, from which it currently collects a rent of € 20 a year. The project will require an initial investment of € 1,000 in furniture which will be depreciated over a period of 4 years, using straight-line depreciation. The additional annual revenues will be of € 900, while the incremental costs of goods sold and other costs are expected to be, respectively, of € 400 (depreciation is not included) and € 50 per year. The value of the working capital is likely to increase by 20% in each year, from year 2 to year 3 and, at the end of the fourth year, the working capital will be 0. The working capital in year 1 would be 500. The tax rate is 38%, the cost of debt is 8.3% (Debt/Assets = 0.7), the risk premium on the firm's stock is 13% and the risk-free rate is 5%. The project duration is 4 years and it is the same as that of the depreciation period. The risk associated with the project and its indebtedness are comparable to that of the firm as a whole.

Calculate NPV and IRR. Verify if the IRR you determined is correct (i.e., NPV = 0).

$$WACC = r_E \cdot E/V + r_D(1-t_c) \cdot D/V = (0,05+0,13) \cdot 0,3 + 0,083 \cdot (1-0,38) \cdot 0,7 = 0,09 = 9\%$$

		Period				
Year		0	1	2	3	4
Panel A Capital Investment						
1	Cash flow from investment in fixed assets	-1000,00				
2	Sales of fixed assets					
3	Less tax on sales					
4	Cash flow from capital investment (1+2+3)	-1000,00				

Panel B Operating Cash Flow							
5	Sales			880,00	880,00	880,00	880,00
6	Cost of goods sold (depreciation not inclu.)			400,00	400,00	400,00	400,00
7	Other costs			50,00	50,00	50,00	50,00
8	Depreciation			250,00	250,00	250,00	250,00
9	Pretax profit (5 - 6 - 7 - 8)			180,00	180,00	180,00	180,00
10	Tax at 38%			68,40	68,40	68,40	68,40
11	Profit after tax (9 - 10)			111,60	111,60	111,60	111,60
12	Operating cash flow (8 + 11)			361,60	361,60	361,60	361,60

Panel C Investment in Working Capital						
13	Working capital		500,00	600,00	720,00	0,00
14	Change in working capital		500,00	100,00	120,00	-720,00
15	Cash flow from investment in w. c . (-14)		-500,00	-100,00	-120,00	720,00

Panel D Project Valuation						
16	Total project cash flow (4 + 12 + 15)	-1000,00	-138,40	261,60	241,60	1.081,60
17	Discount factor at 9%	1,00	0,92	0,84	0,77	0,71
18	Discounted cash flows (16x17)*	-1000,00	-126,97	220,18	186,56	766,23
19	NPV (sum of discounted cash flows)	46,00				

Alternative method for calculating the discounted cash flows (by putting all the data in a single formula)*

Calculations of the discounted cash flows at 9%					
Period	0	1	2	3	4
Formula	$-1000,00/(1+9\%)^0$	$-138,40/(1+9\%)^1$	$261,60/(1+9\%)^2$	$241,60/(1+9\%)^3$	$1.081,60/(1+9\%)^4$
Discounted cash flows	- 1.000,00	-126,97	220,18	186,56	766,23

IRR

IRR = 10,31%

If the IRR is correct, then the NPV at 10.31% will be equal to 0 (in this case it is -0.01 owing to rounding):

Period	0	1	2	3	4
Formula	$-1000/(1+10,31\%)^0$	$138,40/(1+10,31\%)^1$	$261,60/(1+10,31\%)^2$	$241,60/(1+10,31\%)^3$	$1.081,60/(1+10,31\%)^4$
Present values	- 1.000,00	- 125,46	214,98	179,99	730,48
Net present value (sum of PVs)	-0,01				